



```
In [1]: import pandas as pd
import numpy as np
from google.colab import drive
drive.mount('/content/drive')
df = pd.read_csv('/content/drive/My Drive/ADS/supermarket_sales - Sheet1.csv')
```

Mounted at /content/drive

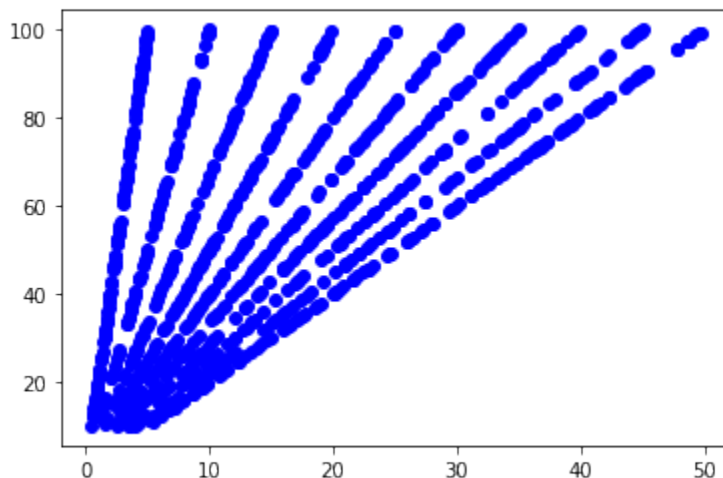
```
In [ ]: df.head()
```

Out[]:

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantit
0	750-67-8428	A	Yangon	Member	Female	Health and beauty	74.69	
1	226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories	15.28	
2	631-41-3108	A	Yangon	Normal	Male	Home and lifestyle	46.33	
3	123-19-1176	A	Yangon	Member	Male	Health and beauty	58.22	
4	373-73-7910	A	Yangon	Normal	Male	Sports and travel	86.31	

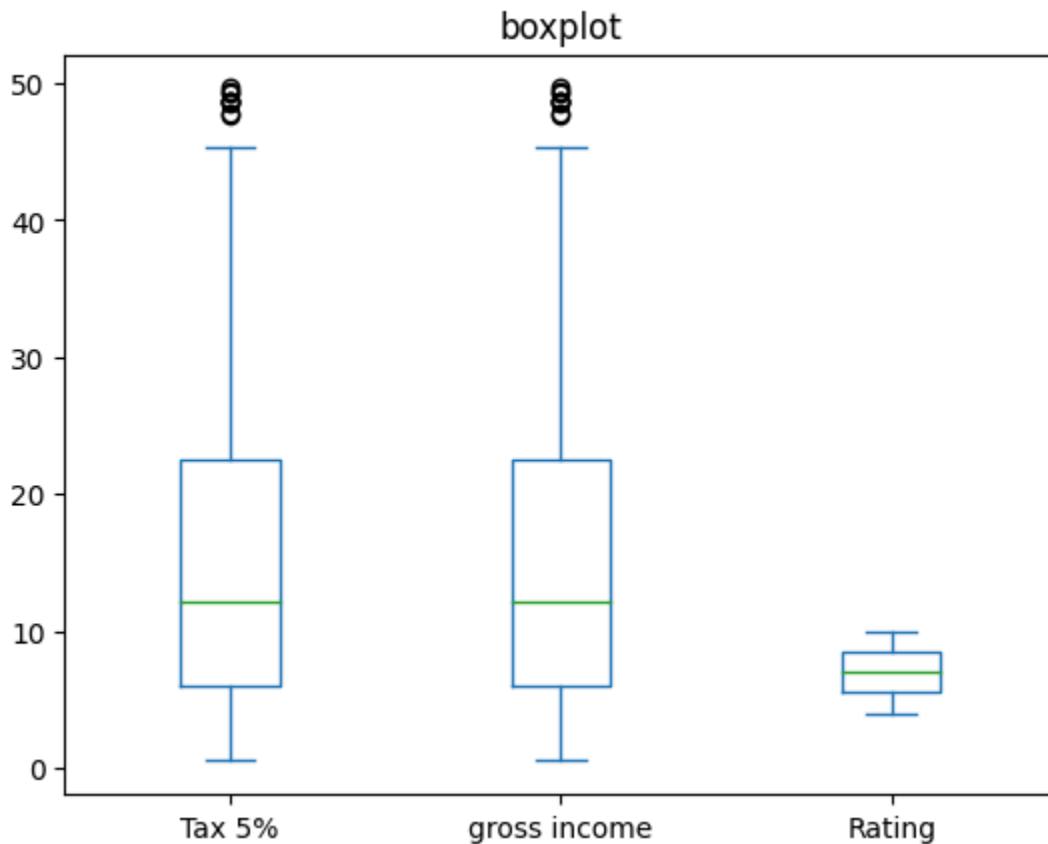
```
In [ ]: #Scatter plot
import matplotlib.pyplot as plt
plt.scatter(df['Tax 5%'], df['Unit price'], c="blue")
```

Out[]: <matplotlib.collections.PathCollection at 0x7f95c022df70>



```
In [9]: #BoxPlot
x2=df['Tax 5%']
x4=df['gross income']
x5=df['Rating']
data = pd.DataFrame({"Tax 5%": x2,"gross income": x4,"Rating": x5})
```

```
# Plot the dataframe
ax = data[['Tax 5%', 'gross income', 'Rating']].plot(kind='box', title='boxplot')
```



```
In [7]: #Distribution Chart / Distplot
import seaborn as sns

g=sns.distplot(df['Total'])
```

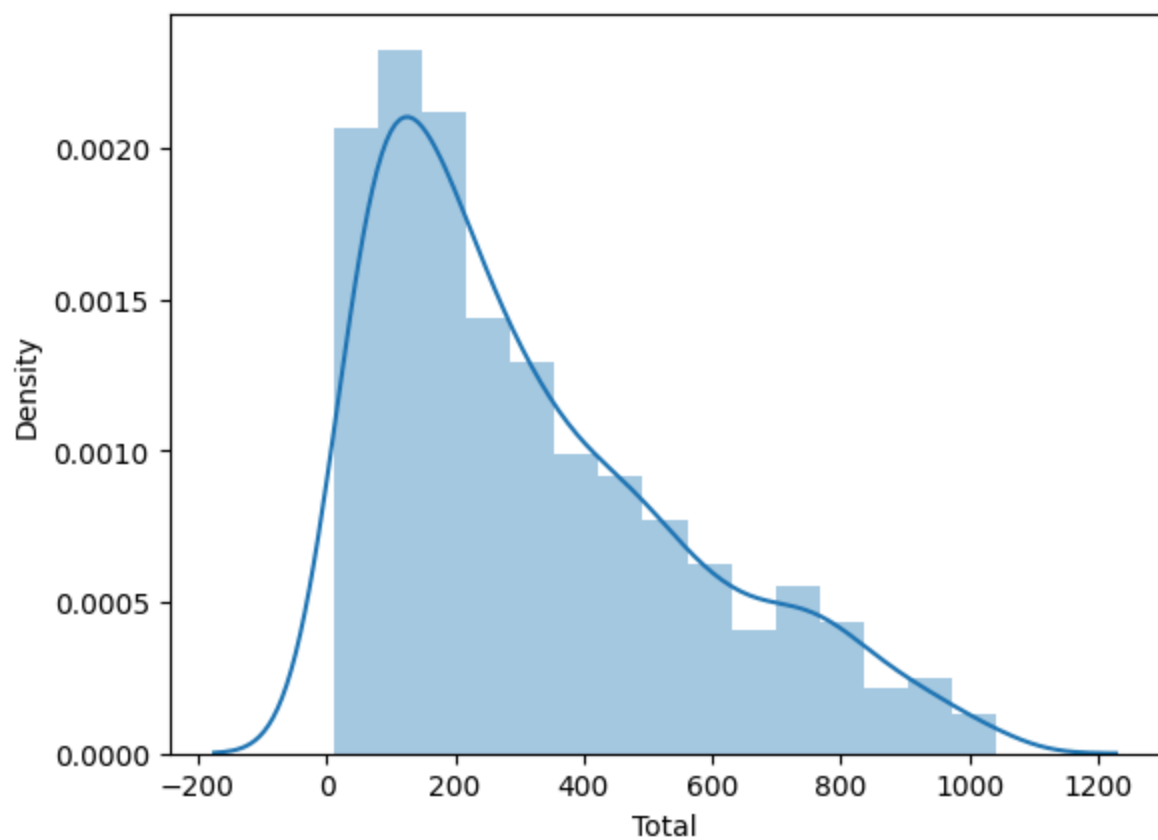
<ipython-input-7-f7ab96f04adf>:4: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

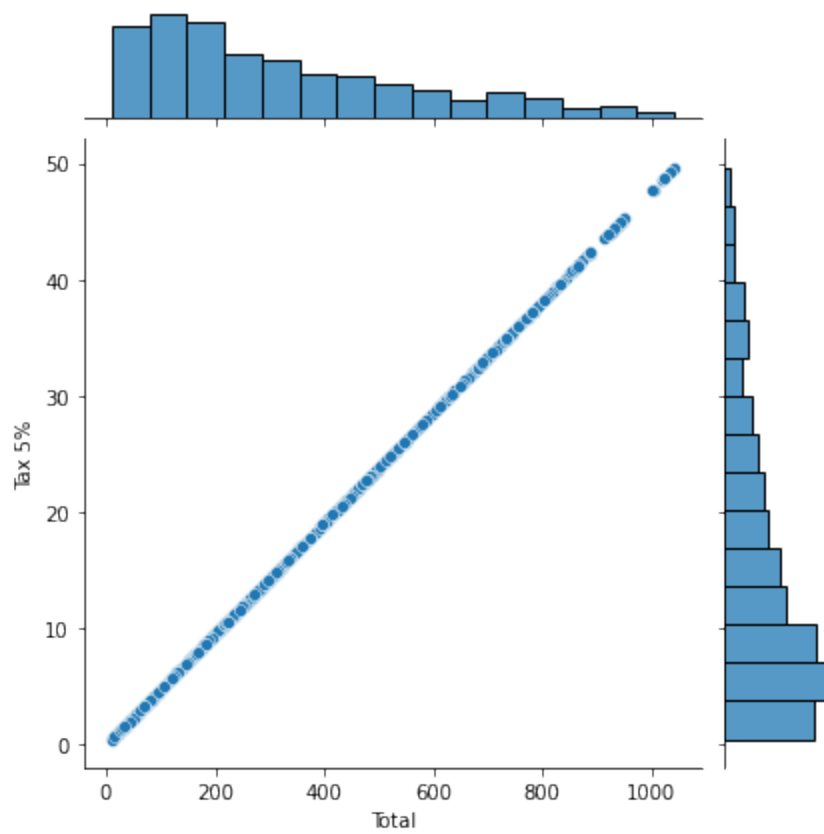
For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
g=sns.distplot(df['Total'])
```

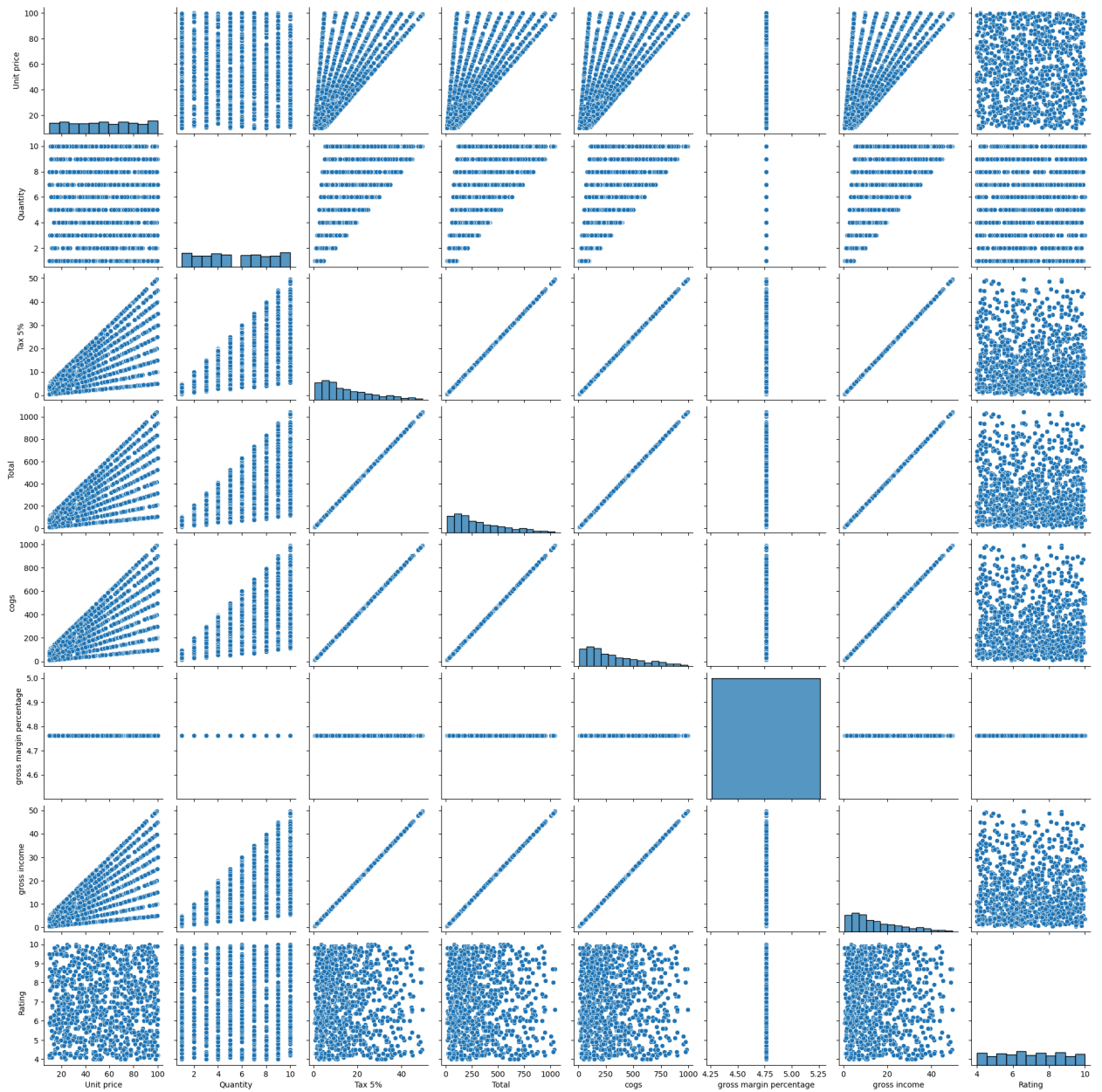


```
In [ ]: #JoinPlot
sns.jointplot(x='Total', y='Tax 5%', data=df)
```

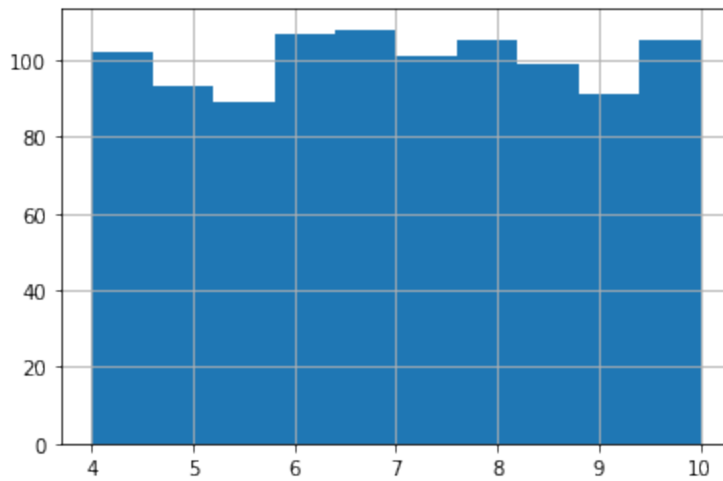
```
Out[ ]: <seaborn.axisgrid.JointGrid at 0x7f95b0e15130>
```



```
In [8]: #Pairplot
sns.pairplot(df)
# to show
plt.show()
```



```
In [ ]: # Histogram
df['Rating'].hist()
plt.show()
```

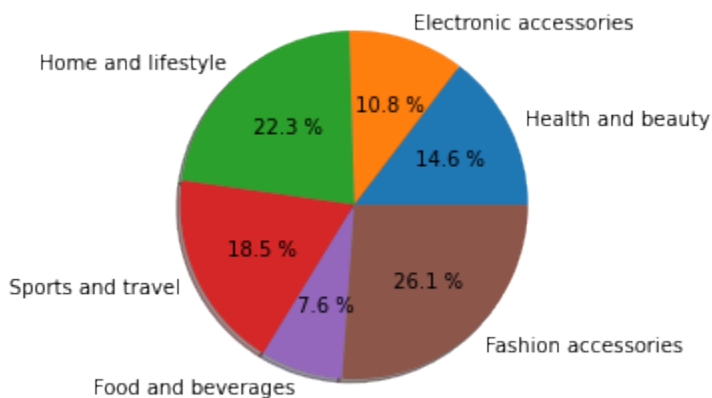


```
In [ ]: lst = df['Product line'].unique()
print(lst)
# t=[0,0,0,0,0,0]
# for i in df:
#     print(i)
#     if i[5] in lst:
#         if i[5]=='Health and beauty':
#             t[0]=t[0]+i[9]

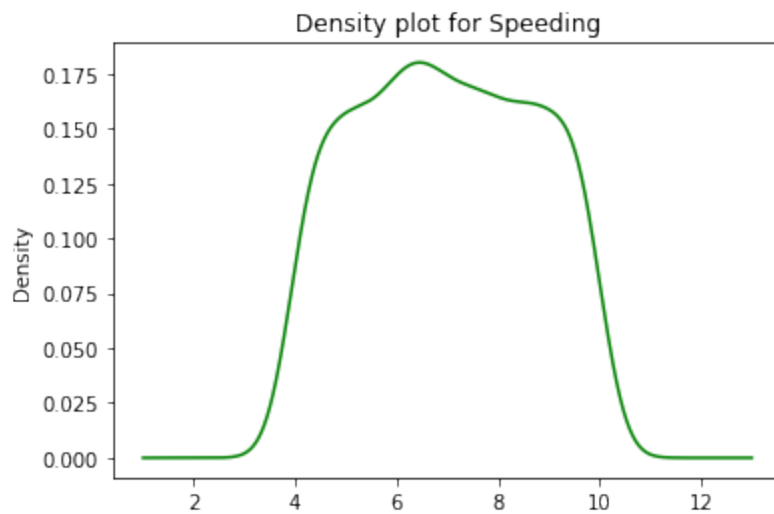
# print(t)

t=[23,17,35,29,12,41]
plt.pie(t,labels=lst,autopct ='% 1.1f %%', shadow = True)
plt.show()
```

```
['Health and beauty' 'Electronic accessories' 'Home and lifestyle'
 'Sports and travel' 'Food and beverages' 'Fashion accessories']
```



```
In [ ]: #Density Chart
df['Rating'].plot.density(color='green')
plt.title('Density plot for Speeding')
plt.show()
```



```
In [ ]: #scatter Matrix
pd.plotting.scatter_matrix(df)
```

```
/usr/local/lib/python3.8/dist-packages/pandas/plotting/_matplotlib/misc.py:100:
UserWarning: Attempting to set identical left == right == 4.761904762 results i
n singular transformations; automatically expanding.
    ax.set_xlim(boundaries_list[j])
/usr/local/lib/python3.8/dist-packages/pandas/plotting/_matplotlib/misc.py:101:
UserWarning: Attempting to set identical bottom == top == 4.761904762 results i
n singular transformations; automatically expanding.
    ax.set_ylim(boundaries_list[i])
/usr/local/lib/python3.8/dist-packages/pandas/plotting/_matplotlib/misc.py:91:
UserWarning: Attempting to set identical left == right == 4.761904762 results i
n singular transformations; automatically expanding.
    ax.set_xlim(boundaries_list[i])
```

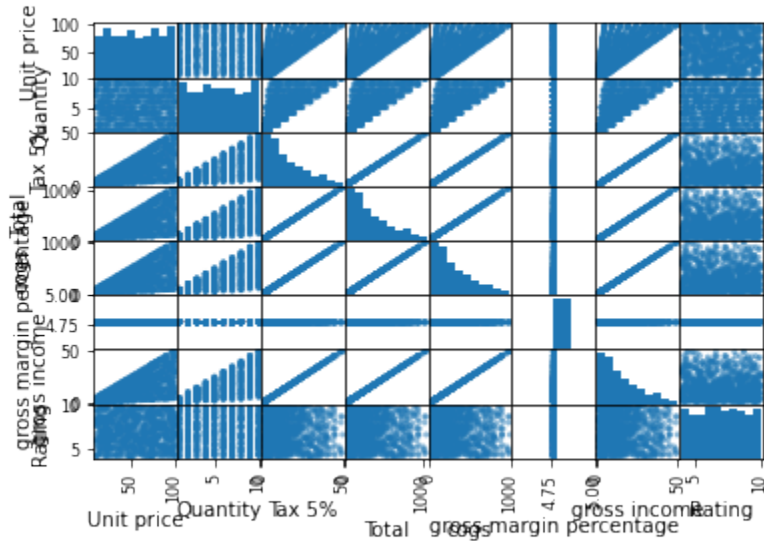
```
Out[ ]: array([[<matplotlib.axes._subplots.AxesSubplot object at 0x7f95c01fa0a0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b62aa310>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b6256730>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b6202b50>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b622ff10>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b61ea310>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b61ea400>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b6195850>],
[<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b6175fa0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b612c3d0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b60db7c0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b6088bb0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b60b6fd0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b6070400>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b601f820>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5fcdc10>],
[<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5ffb0a0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5fb6430>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5f62820>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5f0ec10>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5f3c0a0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5ef6430>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5ea6820>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5e53c10>],
[<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5e020a0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5e3e2b0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5de69d0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5da0130>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5d49850>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5d73f70>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5d286d0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5cd5df0>],
[<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5c8b550>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5cb6c70>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5c6c3d0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5c16af0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5bcd250>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5bf69a0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5ba1160>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5b57850>],
[<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5b03f70>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5b3a6d0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5ae4df0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5a9d550>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5a46c70>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5a7c3d0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5a27af0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b62620d0>],
[<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b60dbeb0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5f8f910>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5e5ffa0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b62de400>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b63b67f0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b63a8c10>],
```



```

<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b842c9d0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5971400>],
[<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b591e7f0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b58cdc40>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b58f80d0>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b58b3520>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b5860910>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b580fd00>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b57c9130>,
<matplotlib.axes._subplots.AxesSubplot object at 0x7f95b57f6520>]],
dtype=object)

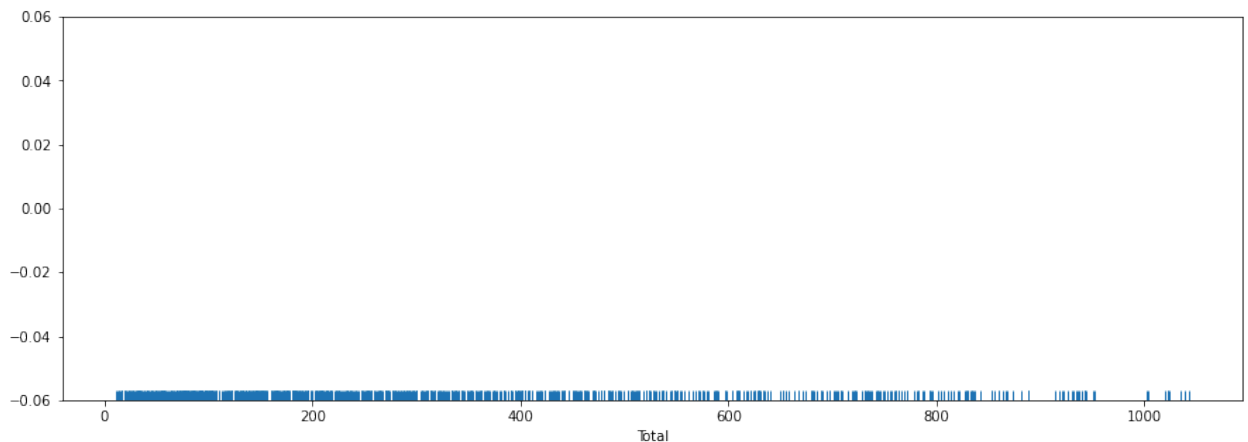
```



```

In [ ]: # rugplot
import seaborn as sns
import matplotlib.pyplot as plt
data = df
data.head(5)
plt.figure(figsize=(15,5))
sns.rugplot(data=data, x="Total")
plt.show()

```

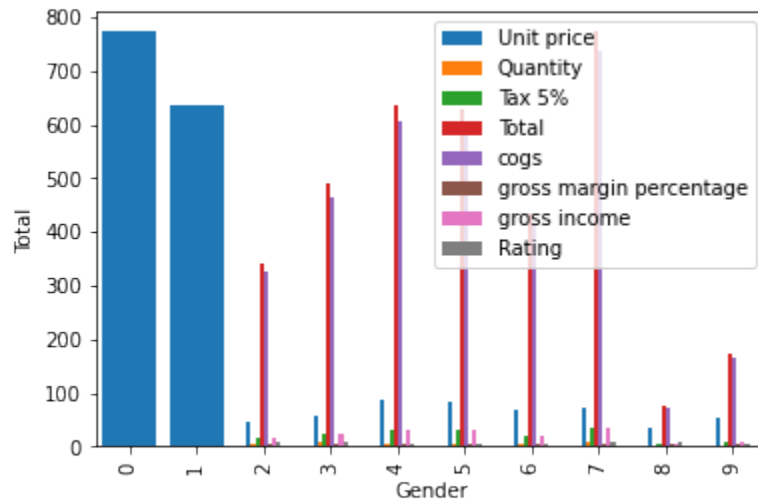


```

In [ ]: #column chart

```

```
# plot between 2 attributes
df1 = df.head(10)
df1.plot.bar()
plt.bar(df1['Gender'], df1['Total'])
plt.xlabel("Gender")
plt.ylabel("Total")
plt.show()
```



```
In [ ]: import plotly.express as px
df1=df.head(15)
fig = px.line(df1, x="Date", y="Total", color='City')
fig.show()
```

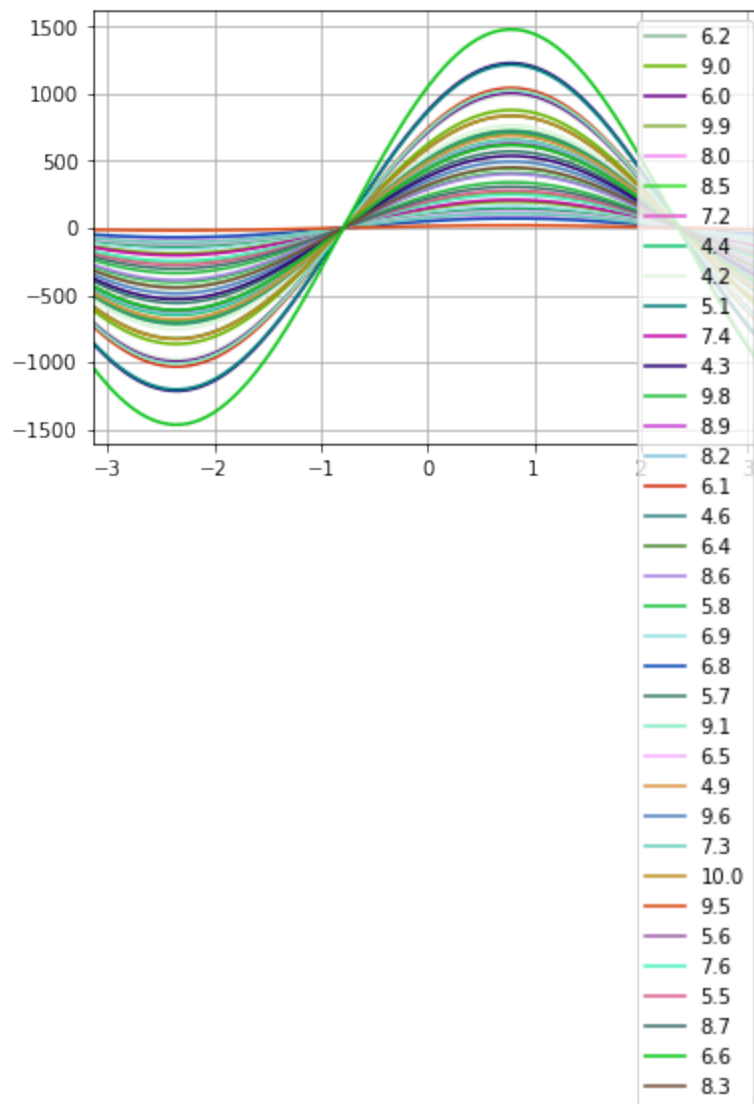
```
In [ ]: #Bubble Chart  
import plotly.express as px  
  
fig = px.scatter(df1, x="Total", y="Tax 5%", size="Quantity", color="City", ho  
fig.show()
```

```
In [ ]: #Parallel
import plotly.express as px
df1 = df.sample(n=100)
fig = px.parallel_coordinates(df1, color="Total",
                             dimensions=['Quantity', 'Unit price', 'Rating'],
                             color_continuous_scale=px.colors.diverging.Tealr,
                             color_continuous_midpoint=2)
fig.show()
```

```
In [ ]: # Creating Andrews curves
df1 = df[['Quantity', 'Total', 'Rating']]
df1=df1.sample(n=50)
x = pd.plotting.andrews_curves(df1, 'Rating')

# plotting the Curve
x.plot()

# Display
plt.show()
```



```
In [ ]: import plotly.express as px

# df = px.data.medals_wide(indexed=True)
fig = px.imshow(df1)
fig.show()
```

```
In [ ]: import plotly.express as px

fig = px.line(df1, x='Date', y="Total")
fig.show()
```

